

Exam-style questions

1 A motor car manufacturer offers various combinations of:

- » seat colours
- » seat materials
- » car paint colours.

A table, CAR, was set up as single-table database to help customers choose which seat and paint combinations were possible. (NOTE: N = no, not a possible combination, Y = yes, combination is possible)

Seat materials
and colours

Paint
colours

Code	Cloth	Leather	Seat colour	White	Red	Black	Blue	Green	Silver	Grey
CB	Y	N	black	Y	Y	Y	Y	Y	Y	Y
LB	N	Y	black	N	Y	N	N	N	Y	Y
CC	Y	N	cream	N	Y	Y	Y	N	N	N
LC	N	Y	cream	N	Y	Y	Y	N	N	Y
CG	Y	N	grey	N	Y	Y	Y	Y	Y	N
LG	N	Y	grey	N	Y	N	Y	N	Y	Y
CR	Y	N	red	Y	N	Y	N	N	Y	Y
LR	N	Y	red	Y	N	Y	N	N	Y	Y
CB	Y	N	blue	N	N	N	Y	N	N	N
LB	N	Y	blue	N	N	Y	Y	Y	N	N

- a i State the number of records shown in the table. [1]
 ii State the number of fields shown in the table. [1]
 iii State, with a reason, if any of the fields are suitable to use as primary key. [2]

b The following SQL command was used:

```
SELECT Code
FROM CAR
WHERE SeatColour = 'Red' AND ('White' OR
'silver');
```

Show what will be displayed. [2]

- c A customer wanted to know all the possible combinations for a car with leather seats and either silver or grey paint colour. Complete the SQL command required.

```
SELECT *
FROM .....
WHERE ..... leather AND (silver ..... grey);
```

[2]

2 A company that sells bicycles keeps records of the items in stock in a table, CYCLE, using a single-table database. For each model of bicycle, the following data is kept:

- model number, for example: BY00007
- description of model, for example: lady's shopper
- colour of frame, for example: gold and black

- size of wheels, for example: 700 mm
- price of model in \$, for example: \$309.50
- still being manufactured: Yes or No
- number in stock.

a i State, with a reason for your choice, the data type you would choose for each field in the table. [7]

ii Identify, with a reason for each, the validation required for each of these fields:

- model number
- price
- number in stock.

[3]

iii State, with a reason, the field that would be suitable to use as primary key. [2]

b SQL commands are used to extract data from the table.

Explain, using examples from your table, how each of these commands could be used:

- **SELECT**
- **FROM**
- **WHERE**
- **SUM**
- **ORDER BY**

[10]

3 A database table, PERFORMANCE, is used to keep a record of the performances at a local theatre.

Show Number	Type	Title	Date	Sold Out
SN091	Comedy	An Evening at Home	01-Sep	Yes
SN102	Drama	Old Places	02-Oct	No
SN113	Jazz	Acoustic Evening	03-Nov	No
SN124	Classical	Mozart Evening	04-Dec	Yes
SN021	Classical	Bach Favourites	01-Feb	Yes
SN032	Jazz	30 Years of Jazz	02-Mar	Yes
SN043	Comedy	Street Night	03-Apr	No
SN054	Comedy	Hoot	04-May	No

a State the number of fields and records in the table.

Fields

Records [2]

b Give **two** validation checks that could be performed on the Show Number field.

Validation check 1

.....

Validation check 2

..... [2]